

Braillix v5.1 - Electronics & Hardware Shopping List

1 Brain Pod + 1 Cell prototype. Multiply cell items by N for more cells.

A. ESP32 Brain Pod

#	Item	Specs	Qty	Price	Source
1	ESP32 DOIT DevKit V1 (30-pin)	CP2102, WiFi+BT, micro-USB	1	Rs 350	Robocraze
2	Female header strip 1x15	2.54mm pitch	2	Rs 10 ea	Local
3	Barrel jack (panel mount)	5.5x2.1mm, threaded	1	Rs 15	Local
4	5V 3A DC adapter	5.5x2.1mm barrel plug	1	Rs 200	Local/Amazon
5	6x6x5mm tactile switch	4-pin DIP, momentary	3	Rs 5 ea	Robocraze
6	4.7k ohm resistor	1/4W through-hole	2	Rs 1 ea	Local
7	Piezo buzzer (optional)	5V, 12mm	1	Rs 15	Local

Pod subtotal: ~Rs 420

B. Per Cell - Motor + Mechanical

#	Item	Specs	Qty	Price	Source
8	28BYJ-48 stepper motor	5V DC, 4-phase, 64:1	1	Rs 80	Robocraze
9	SS49E Hall effect sensor	Linear, SIP-3, 3-6V	1	Rs 25	Robocraze
10	3x2mm NeFeB disc magnet	Neodymium, homing	1	Rs 5	Local
11	2mm SS bearing balls	For braille dot caps	6	Rs 2 ea	Amazon
12	Micro compression spring	OD 4-4.5mm, L 5-6mm	6	Rs 3 ea	Local hw

Per-cell mechanical subtotal: ~Rs 140

C. Per Cell - Electronics (Prototype Route)

#	Item	Specs	Qty	Price	Source
13	Arduino Pro Mini 5V/16MHz	ATmega328P, 33x18mm	1	Rs 120	Robocraze
14	ULN2003 driver board	28x35mm module	1	Rs 40	Robocraze

Per-cell electronics subtotal: ~Rs 160

D. Daisy-Chain Connectors

#	Item	Specs	Qty	Price	Source
15	4-pin pogo connector pair	2mm pitch, spring+pad	2 pairs	Rs 40/pr	LCSC/Alibaba
16	3x2mm NeFeB magnets (docking)	Cell + pod snap faces	9	Rs 5 ea	Local

E. Hardware / Fasteners

#	Item	Specs	Qty	Price	Source
17	M2.5 x 25mm bolts	Pan/socket head	4	Rs 2 ea	Local hw
18	M2 x 6mm screws	Pan head	6	Rs 1 ea	Local hw

#	Item	Specs	Qty	Price	Source
19	M4 bolts + nuts	~10mm, motor mount	2	Rs 2 ea	Local hw
20	M2 grub/set screw	3-4mm (comb, optional)	1	Rs 2	Local hw

F. Wire / Solder Supplies

#	Item	Specs	Qty	Price	Source
21	Hookup wire (stranded)	22-26 AWG, 6 colors	6 rolls	Rs 30/rl	Local
22	Heat shrink tubing	Assorted 1-3mm	1 pack	Rs 30	Local
23	Solder wire	60/40, 0.5-0.8mm	1 roll	Rs 80	Local
24	Flux paste	For SMD soldering	1	Rs 50	Local
25	Superglue (CA)	Balls onto nubs	1 tube	Rs 20	Local
26	Silicone grease	Linkage feet on cam	1 tube	Rs 40	Local

Cost Summary (1 Pod + 1 Cell)

Category	Cost
Brain Pod electronics	Rs 420
Cell mechanical	Rs 140
Cell electronics (Pro Mini)	Rs 160
Connectors + magnets	Rs 125
Hardware/fasteners	Rs 30
Wire/solder/glue	Rs 250
TOTAL (excl. tools + printing)	Rs ~1,125

Quick Counter List (show at shop)

- [] ESP32 DevKit V1 (30-pin) x1
- [] 28BYJ-48 stepper motor x1
- [] Arduino Pro Mini 5V/16MHz x1
- [] ULN2003 driver board x1
- [] SS49E Hall sensor x1
- [] 5V/3A adapter + barrel jack (panel mount) x1
- [] 6x6x5mm tactile switch x3
- [] 4-pin pogo connector pair x2
- [] 3x2mm neodymium magnets x9
- [] 2mm stainless steel bearing balls x6
- [] Micro springs (OD 4mm, length 5mm) x6
- [] Female header 1x15 x2
- [] Male header 40-pin strip x2 (cut to size)
- [] JST XH 5-pin connector x1
- [] Resistors: 4.7k x4, 10k x2
- [] Capacitors: 22pF x2, 100nF x2, 100uF/10V x1
- [] 16MHz crystal x1
- [] M2.5x25 bolts x4
- [] M2x6 screws x6
- [] M4 bolts + nuts x2
- [] Hookup wire (Red, Black, Blue, Yellow, Green, White)
- [] Heat shrink tubing assorted
- [] Solder wire + flux paste
- [] Superglue + silicone grease

Tip: Take calipers to measure the 28BYJ-48 shaft at the shop!

Pogo connectors + bearing balls may need online order (Amazon/Robocraze).